

## Exercise 4 (JOINS)

### Part A (INNER JOIN)

#### Question 1

--Question 1: Show every user who has a subscription. Match users to subscriptions

```
SELECT users.user_id,  
       user_name,  
       subscription_id,  
       start_date  
FROM brightlearn.intelligence.users  
INNER JOIN brightlearn.intelligence.subscriptions  
ON subscriptions.user_id = users.user_id;
```

Add parameter

| Table <span>+</span> <span>Q</span> <span>Y</span> <span>I</span> <span>□</span> <span>brightlearn.intelligence.users</span> <span>⌵</span> <span>^</span> <span>×</span> |                                   |                                     |                                           |                                      |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------|-------------------------------------------|--------------------------------------|--|
|                                                                                                                                                                           | <sup>1</sup> <sub>2</sub> user_id | <sup>A</sup> <sub>C</sub> user_name | <sup>1</sup> <sub>2</sub> subscription_id | <sup>1</sup> <sub>2</sub> start_date |  |
| 1                                                                                                                                                                         | 1                                 | Nomvula                             | 503                                       | 2026-03-10                           |  |
| 2                                                                                                                                                                         | 2                                 | David                               | 502                                       | 2026-02-01                           |  |
| 3                                                                                                                                                                         | 3                                 | Anele                               | 505                                       | 2026-04-05                           |  |
| 4                                                                                                                                                                         | 1                                 | Nomvula                             | 501                                       | 2026-01-15                           |  |

#### Question 2

--Question 2: Show every subscription with its matching plan name and monthly price

```
SELECT subscription_id,  
       user_id,  
       plan_name,  
       monthly_price  
FROM brightlearn.intelligence.subscriptions  
INNER JOIN brightlearn.intelligence.plans  
ON subscriptions.plan_id = plans.plan_id;
```

Add parameter

| Table <span>+</span> <span>Q</span> <span>Y</span> <span>I</span> <span>□</span> <span>brightlearn.intelligence.users</span> <span>⌵</span> <span>^</span> <span>×</span> |                                           |                                   |                                     |                                         |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------|-------------------------------------|-----------------------------------------|--|
|                                                                                                                                                                           | <sup>1</sup> <sub>2</sub> subscription_id | <sup>1</sup> <sub>2</sub> user_id | <sup>A</sup> <sub>C</sub> plan_name | <sup>1</sup> <sub>2</sub> monthly_price |  |
| 1                                                                                                                                                                         | 501                                       | 1                                 | Basic                               | 79                                      |  |
| 2                                                                                                                                                                         | 504                                       | 6                                 | Standard                            | 129                                     |  |
| 3                                                                                                                                                                         | 503                                       | 1                                 | Premium                             | 199                                     |  |
| 4                                                                                                                                                                         | 505                                       | 3                                 | Family                              | 249                                     |  |
| 5                                                                                                                                                                         | 502                                       | 2                                 | Standard                            | 129                                     |  |

### Question 3

```
SELECT session_id,
       user_id,
       show_title,
       genre,
       watch_minutes
FROM brightlearn.intelligence.viewing_sessions
INNER JOIN brightlearn.intelligence.shows
ON viewing_sessions.show_id = shows.show_id;
```

Add parameter

Table  

|   | <sup>1</sup> <sub>2</sub> session_id | <sup>1</sup> <sub>2</sub> show_title | <sup>A</sup> <sub>C</sub> genre | <sup>1</sup> <sub>2</sub> watch_minutes |
|---|--------------------------------------|--------------------------------------|---------------------------------|-----------------------------------------|
| 1 | 904                                  | 7                                    | Comedy                          | 20                                      |
| 2 | 903                                  | 1                                    | Drama                           | 60                                      |
| 3 | 902                                  | 2                                    | Documentary                     | 30                                      |
| 4 | 901                                  | 1                                    | Comedy                          | 45                                      |

### Question 4

--Question 4: Show every viewing session with the user who watched it. Only show sessions with a matching user

```
SELECT user_name,
       city,
       session_id,
       show_id,
       watch_minutes
FROM brightlearn.intelligence.users
INNER JOIN brightlearn.intelligence.viewing_sessions
ON users.user_id = viewing_sessions.user_id;
```

Add parameter

Table  

|   | <sup>A</sup> <sub>C</sub> user_name | <sup>A</sup> <sub>C</sub> city | <sup>1</sup> <sub>2</sub> session_id | <sup>1</sup> <sub>2</sub> show_id | <sup>1</sup> <sub>2</sub> watch_minutes |
|---|-------------------------------------|--------------------------------|--------------------------------------|-----------------------------------|-----------------------------------------|
| 1 | Nomvula                             | Johannesbu...                  | 903                                  | 702                               | 60                                      |
| 2 | David                               | Cape Town                      | 902                                  | 703                               | 30                                      |
| 3 | Anele                               | Durban                         | 905                                  | 705                               | 90                                      |
| 4 | Nomvula                             | Johannesbu...                  | 901                                  | 701                               | 45                                      |

## Question 5

--Question 5: Show users along with their subscriptions, the plan name, and the price. Use only users who have both a subscription and a valid plan

```
SELECT user_name,
       city,
       plan_name,
       monthly_price,
       start_date
FROM brightlearn.intelligence.users
INNER JOIN brightlearn.intelligence.subscriptions
ON users.user_id = subscriptions.user_id
INNER JOIN brightlearn.intelligence.plans
ON subscriptions.plan_id = plans.plan_id;
```

Add parameter

|   | <sup>A</sup> <sub>C</sub> user_name | <sup>A</sup> <sub>C</sub> city | <sup>A</sup> <sub>C</sub> plan_name | <sup>1</sup> <sub>3</sub> monthly_price | <sup>📅</sup> start_date |
|---|-------------------------------------|--------------------------------|-------------------------------------|-----------------------------------------|-------------------------|
| 1 | Nomvula                             | Johannesbu...                  | Premiun                             | 199                                     | 2026-03-10              |
| 2 | David                               | Cape Town                      | Standard                            | 129                                     | 2026-02-01              |
| 3 | Anele                               | Durban                         | Family                              | 249                                     | 2026-04-05              |
| 4 | Nomvula                             | Johannesbu...                  | Basic                               | 79                                      | 2026-01-15              |

## Part B (LEFT JOIN)

## Question 6

--Question 06: Show every user and any subscription they have. Users without subscriptions must still appear.

```
SELECT users.user_id,
       user_name,
       subscription_id,
       start_date
FROM brightlearn.intelligence.users
LEFT JOIN brightlearn.intelligence.subscriptions
```

Add parameter

|   | <sup>1</sup> <sub>3</sub> user_id | <sup>A</sup> <sub>C</sub> user_name | <sup>1</sup> <sub>3</sub> subscription_id | <sup>📅</sup> start_date |
|---|-----------------------------------|-------------------------------------|-------------------------------------------|-------------------------|
| 1 | 1                                 | Nomvula                             | 503                                       | 2026-03-10              |
| 2 | 2                                 | David                               | 502                                       | 2026-02-01              |
| 3 | 3                                 | Anele                               | 505                                       | 2026-04-05              |
| 4 | 4                                 | Kabelo                              | null                                      | null                    |
| 5 | 5                                 | Lerato                              | null                                      | null                    |
| 6 | 1                                 | Nomvula                             | 501                                       | 2026-01-15              |

Question 7

```
--Question 7: Show every plan and the subscriptions on it. Plans with no subscribers must still appear.
SELECT plans.plan_id,
       plan_name,
       subscription_id,
       user_id
FROM   brightlearn.intelligence.plans
LEFT JOIN brightlearn.intelligence.subscriptions
ON plans.plan_id = subscriptions.subscription_id;
```

Add parameter

Table + Q

|    | <sup>1 2</sup> / <sub>3</sub> plan_id | <sup>A B</sup> / <sub>C</sub> plan_name | <sup>1 2</sup> / <sub>3</sub> subscription_id | <sup>1 2</sup> / <sub>3</sub> user_id |
|----|---------------------------------------|-----------------------------------------|-----------------------------------------------|---------------------------------------|
| 1  | 10                                    | Basic                                   | null                                          | null                                  |
| 2  | 11                                    | Standard                                | null                                          | null                                  |
| 3  | 12                                    | Premiun                                 | null                                          | null                                  |
| 4  | 13                                    | Family                                  | null                                          | null                                  |
| 5  | 14                                    | Mobile                                  | null                                          | null                                  |
| 6  | 10                                    | Basic                                   | null                                          | null                                  |
| 7  | 11                                    | Standard                                | null                                          | null                                  |
| 8  | 12                                    | Premiun                                 | null                                          | null                                  |
| 9  | 13                                    | Family                                  | null                                          | null                                  |
| 10 | 14                                    | Mobile                                  | null                                          | null                                  |

## Question 8

--Question 8: Show every show and any viewing sessions on it. Shows that were never watched must still appear

```
SELECT shows.show_id,
       show_title,
       session_id,
       watch_minutes
FROM brightlearn.intelligence.shows
LEFT JOIN brightlearn.intelligence.viewing_sessions
ON shows.show_id = viewing_sessions.show_id;
```

Table  

|    | <sup>1</sup> <sub>3</sub> show_id | <sup>A</sup> <sub>C</sub> show_title | <sup>1</sup> <sub>3</sub> session_id | <sup>1</sup> <sub>3</sub> watch_minutes |
|----|-----------------------------------|--------------------------------------|--------------------------------------|-----------------------------------------|
| 1  | 701                               | Comedy Hour                          | 904                                  | 20                                      |
| 2  | 702                               | Crime Time                           | 903                                  | 60                                      |
| 3  | 703                               | Tech Tales                           | 902                                  | 30                                      |
| 4  | 704                               | Cooking Lab                          | null                                 | null                                    |
| 5  | 706                               | Wild Earth                           | null                                 | null                                    |
| 6  | 701                               | Comedy Hour                          | 901                                  | 45                                      |
| 7  | 702                               | Crime Time                           | 903                                  | 60                                      |
| 8  | 703                               | Tech Tales                           | 902                                  | 30                                      |
| 9  | 701                               | Comedy Hour                          | 904                                  | 20                                      |
| 10 | 701                               | Comedy Hour                          | 901                                  | 45                                      |
| 11 | 701                               | Comedy Hour                          | 904                                  | 20                                      |
| 12 | 702                               | Crime Time                           | 903                                  | 60                                      |
| 13 | 703                               | Tech Tales                           | 902                                  | 30                                      |
| 14 | 704                               | Cooking Lab                          | null                                 | null                                    |
| 15 | 706                               | Wild Earth                           | null                                 | null                                    |
| 16 | 701                               | Comedy Hour                          | 901                                  | 45                                      |
| 17 | 702                               | Crime Time                           | 903                                  | 60                                      |
| 18 | 703                               | Tech Tales                           | 902                                  | 30                                      |
| 19 | 701                               | Comedy Hour                          | 904                                  | 20                                      |
| 20 | 701                               | Comedy Hour                          | 901                                  | 45                                      |

## Question 9

--Question 9: Show every viewing session and the user who watched it. Sessions referencing users that do not exist must still appear (with NULL user details)

```
SELECT viewing_sessions.session_id,
       show_id,
       watch_minutes,
       viewing_sessions.user_id,
       user_name
FROM brightlearn.intelligence.viewing_sessions
LEFT JOIN brightlearn.intelligence.users
ON viewing_sessions.user_id = users.user_id;
```

|    | 1 <sup>2</sup> <sub>3</sub> session_id | 1 <sup>2</sup> <sub>3</sub> show_id | 1 <sup>2</sup> <sub>3</sub> watch_minutes | 1 <sup>2</sup> <sub>3</sub> user_id | A <sup>B</sup> <sub>C</sub> user_name |
|----|----------------------------------------|-------------------------------------|-------------------------------------------|-------------------------------------|---------------------------------------|
| 1  | 901                                    | 701                                 | 45                                        | 1                                   | Nomvula                               |
| 2  | 902                                    | 703                                 | 30                                        | 2                                   | David                                 |
| 3  | 903                                    | 702                                 | 60                                        | 1                                   | Nomvula                               |
| 4  | 904                                    | 701                                 | 20                                        | 7                                   | null                                  |
| 5  | 905                                    | 705                                 | 90                                        | 3                                   | Anele                                 |
| 6  | 901                                    | 701                                 | 45                                        | 1                                   | Nomvula                               |
| 7  | 902                                    | 703                                 | 30                                        | 2                                   | David                                 |
| 8  | 903                                    | 702                                 | 60                                        | 1                                   | Nomvula                               |
| 9  | 905                                    | 705                                 | 90                                        | 3                                   | Anele                                 |
| 10 | 901                                    | 701                                 | 45                                        | 1                                   | Nomvula                               |
| 11 | 902                                    | 703                                 | 30                                        | 2                                   | David                                 |
| 12 | 903                                    | 702                                 | 60                                        | 1                                   | Nomvula                               |
| 13 | 904                                    | 701                                 | 20                                        | 7                                   | null                                  |
| 14 | 905                                    | 705                                 | 90                                        | 3                                   | Anele                                 |
| 15 | 901                                    | 701                                 | 45                                        | 1                                   | Nomvula                               |
| 16 | 902                                    | 703                                 | 30                                        | 2                                   | David                                 |
| 17 | 903                                    | 702                                 | 60                                        | 1                                   | Nomvula                               |
| 18 | 905                                    | 705                                 | 90                                        | 3                                   | Anele                                 |

## Question 10

--Question 9: Show every viewing session and the user who watched it. Sessions referencing users that do not exist must still appear (with NULL user details)

```
SELECT viewing_sessions.session_id,
       show_id,
       watch_minutes,
       viewing_sessions.user_id,
       user_name
FROM brightlearn.intelligence.viewing_sessions
LEFT JOIN brightlearn.intelligence.users
ON viewing_sessions.user_id = users.user_id;
```

|    | user_name | city           | plan_name | monthly_price |
|----|-----------|----------------|-----------|---------------|
| 1  | Nomvula   | Johannesbu...  | Premiun   | 199           |
| 2  | David     | Cape Town      | Standard  | 129           |
| 3  | Anele     | Durban         | Family    | 249           |
| 4  | Kabelo    | Pretoria       | null      | null          |
| 5  | Lerato    | Port Elizabeth | null      | null          |
| 6  | Nomvula   | Johannesbu...  | Basic     | 79            |
| 7  | David     | Cape Town      | Standard  | 129           |
| 8  | Anele     | Durban         | Family    | 249           |
| 9  | Nomvula   | Johannesbu...  | Premiun   | 199           |
| 10 | Nomvula   | Johannesbu...  | Basic     | 79            |
| 11 | Nomvula   | Johannesbu...  | Premiun   | 199           |
| 12 | David     | Cape Town      | Standard  | 129           |
| 13 | Anele     | Durban         | Family    | 249           |
| 14 | Nomvula   | Johannesbu...  | Basic     | 79            |
| 15 | David     | Cape Town      | Standard  | 129           |
| 16 | Anele     | Durban         | Family    | 249           |
| 17 | Nomvula   | Johannesbu...  | Premiun   | 199           |
| 18 | Nomvula   | Johannesbu...  | Basic     | 79            |
| 19 | Nomvula   | Johannesbu...  | Premiun   | 199           |
| 20 | David     | Cape Town      | Standard  | 129           |
| 21 | Anele     | Durban         | Family    | 249           |
| 22 | Kabelo    | Pretoria       | null      | null          |
| 23 | Lerato    | Port Elizabeth | null      | null          |

|    |         |               |          |     |
|----|---------|---------------|----------|-----|
| 24 | Nomvula | Johannesbu... | Basic    | 79  |
| 25 | David   | Cape Town     | Standard | 129 |
| 26 | Anele   | Durban        | Family   | 249 |
| 27 | Nomvula | Johannesbu... | Premiun  | 199 |
| 28 | Nomvula | Johannesbu... | Basic    | 79  |
| 29 | Nomvula | Johannesbu... | Premiun  | 199 |
| 30 | David   | Cape Town     | Standard | 129 |
| 31 | Anele   | Durban        | Family   | 249 |
| 32 | Nomvula | Johannesbu... | Basic    | 79  |
| 33 | David   | Cape Town     | Standard | 129 |
| 34 | Anele   | Durban        | Family   | 249 |

|    |         |               |         |     |
|----|---------|---------------|---------|-----|
| 35 | Nomvula | Johannesbu... | Premiun | 199 |
| 36 | Nomvula | Johannesbu... | Basic   | 79  |

Part C (FULL OUTER JOIN)

Question 11

```
--Question 11: Show every user and every subscription, including users without subscriptions AND subscriptions referencing users that do not exist

SELECT users.user_id,
       user_name,
       subscription_id,
       start_date
FROM brightlearn.intelligence.users
FULL OUTER JOIN brightlearn.intelligence.subscriptions
ON users.user_id = subscriptions.subscription_id;
```

| Table ▾ + |             |               |                     |              |
|-----------|-------------|---------------|---------------------|--------------|
|           | 123 user_id | A8C user_name | 123 subscription_id | 📅 start_date |
| 1         | 5           | Lerato        | null                | null         |
| 2         | 1           | Nomvula       | null                | null         |
| 3         | 3           | Anele         | null                | null         |
| 4         | 2           | David         | null                | null         |
| 5         | 4           | Kabelo        | null                | null         |
| 6         | 5           | Lerato        | null                | null         |
| 7         | 1           | Nomvula       | null                | null         |
| 8         | 3           | Anele         | null                | null         |
| 9         | 2           | David         | null                | null         |
| 10        | 4           | Kabelo        | null                | null         |
| 11        | null        | null          | 504                 | 2026-03-20   |
| 12        | null        | null          | 504                 | 2026-03-20   |
| 13        | null        | null          | 501                 | 2026-01-15   |
| 14        | null        | null          | 501                 | 2026-01-15   |
| 15        | null        | null          | 503                 | 2026-03-10   |
| 16        | null        | null          | 503                 | 2026-03-10   |
| 17        | null        | null          | 502                 | 2026-02-01   |
| 18        | null        | null          | 502                 | 2026-02-01   |
| 19        | null        | null          | 505                 | 2026-04-05   |
| 20        | null        | null          | 505                 | 2026-04-05   |

## Question 12

--Question 12: Show every plan and every subscription, including plans without subscribers AND any subscription referencing a plan that does not exist

```

SELECT plans.plan_id,
       plan_name,
       subscription_id,
       user_id
FROM brightlearn.intelligence.plans
FULL OUTER JOIN brightlearn.intelligence.subscriptions
ON plans.plan_id = subscriptions.plan_id;

```

|    | plan_id | plan_name | subscription_id | user_id |
|----|---------|-----------|-----------------|---------|
| 1  | 12      | Premium   | 503             | 1       |
| 2  | 10      | Basic     | 501             | 1       |
| 3  | 13      | Family    | 505             | 3       |
| 4  | 14      | Mobile    | null            | null    |
| 5  | 11      | Standard  | 504             | 6       |
| 6  | 12      | Premium   | 503             | 1       |
| 7  | 10      | Basic     | 501             | 1       |
| 8  | 13      | Family    | 505             | 3       |
| 9  | 14      | Mobile    | null            | null    |
| 10 | 11      | Standard  | 504             | 6       |
| 11 | 12      | Premium   | 503             | 1       |
| 12 | 10      | Basic     | 501             | 1       |
| 13 | 13      | Family    | 505             | 3       |
| 14 | 11      | Standard  | 502             | 2       |
| 15 | 12      | Premium   | 503             | 1       |
| 16 | 10      | Basic     | 501             | 1       |
| 17 | 13      | Family    | 505             | 3       |
| 18 | 11      | Standard  | 502             | 2       |
| 19 | 11      | Standard  | 504             | 6       |
| 20 | 11      | Standard  | 504             | 6       |
| 21 | 11      | Standard  | 502             | 2       |
| 22 | 11      | Standard  | 502             | 2       |

### Question 13

```
--Question 13: Show every show and every viewing session, including shows that were never watched AND sessions referencing shows that do not exist
SELECT shows.show_id,
       show_title,
       session_id,
       watch_minutes
FROM brightlearn.intelligence.shows
FULL OUTER JOIN brightlearn.intelligence.viewing_sessions
ON shows.show_id = viewing_sessions.show_id;
```

|    | 1 <sup>2</sup> <sub>3</sub> show_id | 1 <sup>2</sup> <sub>3</sub> show_title | 1 <sup>2</sup> <sub>3</sub> session_id | 1 <sup>2</sup> <sub>3</sub> watch_minutes |
|----|-------------------------------------|----------------------------------------|----------------------------------------|-------------------------------------------|
| 1  | 704                                 | Cooking Lab                            | null                                   | null                                      |
| 2  | 703                                 | Tech Tales                             | 902                                    | 30                                        |
| 3  | 701                                 | Comedy Hour                            | 904                                    | 20                                        |
| 4  | 702                                 | Crime Time                             | 903                                    | 60                                        |
| 5  | 706                                 | Wild Earth                             | null                                   | null                                      |
| 6  | 704                                 | Cooking Lab                            | null                                   | null                                      |
| 7  | 703                                 | Tech Tales                             | 902                                    | 30                                        |
| 8  | 701                                 | Comedy Hour                            | 904                                    | 20                                        |
| 9  | 702                                 | Crime Time                             | 903                                    | 60                                        |
| 10 | 706                                 | Wild Earth                             | null                                   | null                                      |
| 11 | 703                                 | Tech Tales                             | 902                                    | 30                                        |
| 12 | 701                                 | Comedy Hour                            | 901                                    | 45                                        |
| 13 | 702                                 | Crime Time                             | 903                                    | 60                                        |
| 14 | 703                                 | Tech Tales                             | 902                                    | 30                                        |
| 15 | 701                                 | Comedy Hour                            | 901                                    | 45                                        |
| 16 | 702                                 | Crime Time                             | 903                                    | 60                                        |
| 17 | 701                                 | Comedy Hour                            | 904                                    | 20                                        |
| 18 | 701                                 | Comedy Hour                            | 904                                    | 20                                        |
| 19 | 701                                 | Comedy Hour                            | 901                                    | 45                                        |
| 20 | 701                                 | Comedy Hour                            | 901                                    | 45                                        |
| 21 | null                                | null                                   | 905                                    | 90                                        |
| 22 | null                                | null                                   | 905                                    | 90                                        |

## Question 14

✓ --Question 14: Show every user and every viewing session, including users with no sessions AND sessions referencing users who do not exist

```
✓ SELECT users.user_id,
       user_name,
       session_id,
       show_id,
       watch_minutes
FROM brightlearn.intelligence.users
FULL OUTER JOIN brightlearn.intelligence.viewing_sessions
ON users.user_id = viewing_sessions.user_id;
```

Table  

|    | <sup>1</sup> <sub>3</sub> user_id | <sup>A</sup> <sub>C</sub> user_name | <sup>1</sup> <sub>3</sub> session_id | <sup>1</sup> <sub>3</sub> show_id | <sup>1</sup> <sub>3</sub> watch_minutes |
|----|-----------------------------------|-------------------------------------|--------------------------------------|-----------------------------------|-----------------------------------------|
| 1  | 1                                 | Nomvula                             | 901                                  | 701                               | 45                                      |
| 2  | 1                                 | Nomvula                             | 903                                  | 702                               | 60                                      |
| 3  | 3                                 | Anele                               | 905                                  | 705                               | 90                                      |
| 4  | 2                                 | David                               | 902                                  | 703                               | 30                                      |
| 5  | null                              | null                                | 904                                  | 701                               | 20                                      |
| 6  | 1                                 | Nomvula                             | 901                                  | 701                               | 45                                      |
| 7  | 1                                 | Nomvula                             | 903                                  | 702                               | 60                                      |
| 8  | 3                                 | Anele                               | 905                                  | 705                               | 90                                      |
| 9  | 2                                 | David                               | 902                                  | 703                               | 30                                      |
| 10 | null                              | null                                | 904                                  | 701                               | 20                                      |
| 11 | 1                                 | Nomvula                             | 901                                  | 701                               | 45                                      |
| 12 | 1                                 | Nomvula                             | 903                                  | 702                               | 60                                      |

|    |   |         |      |      |      |
|----|---|---------|------|------|------|
| 13 | 3 | Anele   | 905  | 705  | 90   |
| 14 | 2 | David   | 902  | 703  | 30   |
| 15 | 1 | Nomvula | 901  | 701  | 45   |
| 16 | 1 | Nomvula | 903  | 702  | 60   |
| 17 | 3 | Anele   | 905  | 705  | 90   |
| 18 | 2 | David   | 902  | 703  | 30   |
| 19 | 4 | Kabelo  | null | null | null |
| 20 | 4 | Kabelo  | null | null | null |
| 21 | 5 | Lerato  | null | null | null |
| 22 | 5 | Lerato  | null | null | null |

## Question 15

--Question 15: Show every user, every subscription and every plan in one query - using FULL OUTER JOIN throughout. This is the hardest question - get all gaps visible at once

```
SELECT users.user_id,
       user_name,
       subscription_id,
       plans.plan_id,
       plan_name
FROM brightlearn.intelligence.users
FULL OUTER JOIN brightlearn.intelligence.subscriptions
ON users.user_id = subscriptions.user_id
FULL OUTER JOIN brightlearn.intelligence.plans
ON subscriptions.plan_id = plans.plan_id;
```

|    | 1 <sup>2</sup> <sub>3</sub> user_id | A <sup>B</sup> <sub>C</sub> user_name | 1 <sup>2</sup> <sub>3</sub> subscription_id | 1 <sup>2</sup> <sub>3</sub> plan_id | A <sup>B</sup> <sub>C</sub> plan_name |
|----|-------------------------------------|---------------------------------------|---------------------------------------------|-------------------------------------|---------------------------------------|
| 1  | 1                                   | Nomvula                               | 503                                         | 12                                  | Premiun                               |
| 2  | 1                                   | Nomvula                               | 503                                         | 12                                  | Premiun                               |
| 3  | 1                                   | Nomvula                               | 503                                         | 12                                  | Premiun                               |
| 4  | 1                                   | Nomvula                               | 503                                         | 12                                  | Premiun                               |
| 5  | 1                                   | Nomvula                               | 501                                         | 10                                  | Basic                                 |
| 6  | 1                                   | Nomvula                               | 501                                         | 10                                  | Basic                                 |
| 7  | 1                                   | Nomvula                               | 501                                         | 10                                  | Basic                                 |
| 8  | 1                                   | Nomvula                               | 501                                         | 10                                  | Basic                                 |
| 9  | 3                                   | Anele                                 | 505                                         | 13                                  | Family                                |
| 10 | 3                                   | Anele                                 | 505                                         | 13                                  | Family                                |
| 11 | 3                                   | Anele                                 | 505                                         | 13                                  | Family                                |
| 12 | 3                                   | Anele                                 | 505                                         | 13                                  | Family                                |
| 13 | 5                                   | Lerato                                | null                                        | null                                | null                                  |
| 14 | 4                                   | Kabelo                                | null                                        | null                                | null                                  |
| 15 | 5                                   | Lerato                                | null                                        | null                                | null                                  |
| 16 | 4                                   | Kabelo                                | null                                        | null                                | null                                  |
| 17 | 2                                   | David                                 | 502                                         | 11                                  | Standard                              |
| 18 | 2                                   | David                                 | 502                                         | 11                                  | Standard                              |
| 19 | 2                                   | David                                 | 502                                         | 11                                  | Standard                              |
| 20 | 2                                   | David                                 | 502                                         | 11                                  | Standard                              |
| 21 | null                                | null                                  | 504                                         | 11                                  | Standard                              |
| 22 | null                                | null                                  | 504                                         | 11                                  | Standard                              |
| 23 | 1                                   | Nomvula                               | 503                                         | 12                                  | Premiun                               |
| 24 | 1                                   | Nomvula                               | 503                                         | 12                                  | Premiun                               |

|    |   |         |     |    |          |
|----|---|---------|-----|----|----------|
| 25 | 1 | Nomvula | 503 | 12 | Premiun  |
| 26 | 1 | Nomvula | 503 | 12 | Premiun  |
| 27 | 1 | Nomvula | 501 | 10 | Basic    |
| 28 | 1 | Nomvula | 501 | 10 | Basic    |
| 29 | 1 | Nomvula | 501 | 10 | Basic    |
| 30 | 1 | Nomvula | 501 | 10 | Basic    |
| 31 | 3 | Anele   | 505 | 13 | Family   |
| 32 | 3 | Anele   | 505 | 13 | Family   |
| 33 | 3 | Anele   | 505 | 13 | Family   |
| 34 | 3 | Anele   | 505 | 13 | Family   |
| 35 | 2 | David   | 502 | 11 | Standard |

|    |      |       |      |    |          |
|----|------|-------|------|----|----------|
| 36 | 2    | David | 502  | 11 | Standard |
| 37 | 2    | David | 502  | 11 | Standard |
| 38 | 2    | David | 502  | 11 | Standard |
| 39 | null | null  | 504  | 11 | Standard |
| 40 | null | null  | 504  | 11 | Standard |
| 41 | null | null  | null | 14 | Mobile   |
| 42 | null | null  | null | 14 | Mobile   |

## BONUS CHALLENGE

### Bonus 1

--Bonus 1: Which users have not subscribed to any plan

```
SELECT user_name
FROM brightlearn.intelligence.users
LEFT JOIN brightlearn.intelligence.subscriptions
ON users.user_id = subscriptions.user_id
WHERE subscription_id IS NULL;
```

Add parameter

Table  

|   |   user_name |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Kabelo                                                                                                                                                                        |
| 2 | Lerato                                                                                                                                                                        |
| 3 | Kabelo                                                                                                                                                                        |
| 4 | Lerato                                                                                                                                                                        |

### Bonus 2

--Bonus 2: Which subscriptions reference users that do not exist in the users table

```
SELECT subscription_id
FROM brightlearn.intelligence.subscriptions
LEFT JOIN brightlearn.intelligence.users
ON subscriptions.user_id = users.user_id
WHERE users.user_id IS NULL;
```

Add parameter

Table  

      

|   |   subscription_id |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | 504                                                                                                                                                                                     |
| 2 | 504                                                                                                                                                                                     |

### Bonus 3

```
--Bonus 3: Which shows have never been watched

SELECT show_title,
       shows.show_id
FROM brightlearn.intelligence.shows
LEFT JOIN brightlearn.intelligence.viewing_sessions
ON shows.show_id = viewing_sessions.show_id
WHERE session_id IS NULL;
```

Add parameter

Table  

|   | <sup>A</sup> <sub>C</sub> show_title | <sup>1</sup> <sub>3</sub> show_id |
|---|--------------------------------------|-----------------------------------|
| 1 | Cooking Lab                          | 704                               |
| 2 | Wild Earth                           | 706                               |
| 3 | Cooking Lab                          | 704                               |
| 4 | Wild Earth                           | 706                               |

### Bonus 4

```
--Bonus 4: Which viewing sessions reference shows that do not exist

SELECT session_id,
       viewing_sessions.show_id
FROM brightlearn.intelligence.viewing_sessions
LEFT JOIN brightlearn.intelligence.shows
ON viewing_sessions.show_id = shows.show_id
WHERE shows.show_id IS NULL;
```

Add parameter

Table  

|   | <sup>1</sup> <sub>3</sub> session_id | <sup>1</sup> <sub>3</sub> show_id |
|---|--------------------------------------|-----------------------------------|
| 1 | 905                                  | 705                               |
| 2 | 905                                  | 705                               |

## Bonus 5

```
--Bonus 5: Which plans have no subscribers
SELECT plan_name,
       plans.plan_id
FROM brightlearn.intelligence.plans
LEFT JOIN brightlearn.intelligence.subscriptions
ON plans.plan_id = subscriptions.plan_id
WHERE subscription_id IS NULL;
```

Add parameter

Table  

|   |  plan_name |  plan_id |
|---|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 1 | Mobile                                                                                      | 14                                                                                        |
| 2 | Mobile                                                                                      | 14                                                                                        |